Input system:
Dependent variables:y, x
Independent variables:t

$$\frac{dx}{dt} = ax - bxy\text{inv}((1 + Ax)(1 + By))$$

$$\frac{dy}{dt} = (-1)cy + dxy\text{inv}((1 + Ax)(1 + By))$$

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Result:
Dependent variables:y, s_1, x
Independent variables:t
Additional variables:

$$s_1 = \text{inv}((1 + Ax)(1 + By))$$

$$\frac{dx}{dt} = (-1)bxys_1 + ax$$

$$\frac{dy}{dt} = dxys_1 + (-1)cy$$

$$\frac{ds_1}{dt} = s_1^3By^2Abx+(-1)s_1^3Ax^2Bdy+s_1^3Abxy+(-1)s_1^2ByAax+(-1)s_1^3Bdxy+s_1^2AxBcy+(-1)s_1^2Aax+s_1^2Bcy$$